

Topic 4: GitHub Copilot

4.1 What is GitHub Copilot?

- GitHub Copilot is an AI coding assistant that integrates directly into your code editor
- Built by GitHub (Microsoft) in partnership with OpenAI, trained on billions of lines of public code
- Integrates with: VS Code, JetBrains IDEs (IntelliJ, PyCharm, etc.), Neovim, Visual Studio, Azure Data Studio
- Pricing: free for students and open-source maintainers; individual \$10/month; Business \$19/user/month
- Documentation: <https://github.com/features/copilot>

4.2 Core Features

- Inline completions: as you type, Copilot suggests the next line or block of code -- press Tab to accept
- Copilot Chat: ask questions in natural language directly in your editor ('Explain this function', 'Fix this bug')
- Generate from comments: write a plain-English comment describing the function, Copilot writes the implementation
- Terminal integration: Copilot explains shell commands and suggests fixes for command errors
- Pull request descriptions: automatically generates PR summaries from your code diff

```
# Write a comment, Copilot generates the implementation:

# Read a CSV file and return only rows where
# the 'status' column equals 'active'
def get_active_users(filepath: str) -> list[dict]:
    import csv
    with open(filepath, newline='') as f:
        reader = csv.DictReader(f)
        return [row for row in reader if row['status'] == 'active']
```

4.3 Getting the Most from Copilot

- Write descriptive comments before functions -- Copilot uses them as instructions for the implementation
- Name functions and variables clearly -- good names guide better and more relevant completions
- Use Copilot Chat for explanation: select a confusing block of code, right-click, and ask 'Explain this'

- Generate tests: open a function file and ask Copilot Chat to 'Write unit tests for this module'
- Always review every suggestion -- Copilot can introduce subtle bugs, especially around edge cases and security

4.4 Alternatives to Copilot

- Cursor (cursor.com): AI-first code editor (fork of VS Code) with deeper AI integration; free and paid tiers
 - Codeium (codeium.com): free alternative to Copilot; good for personal and student projects
 - Amazon CodeWhisperer: free for individuals; well-suited to AWS-related code
 - Tabnine (tabnine.com): privacy-focused; enterprise option allows models to run locally
 - Continue.dev: open-source, connects to any LLM including locally-run Ollama models
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